Colton Feathers

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Education

- **Ohio University** – Athens, OH USA
  - *Degree*: B.S. Astrophysics, Mathematics Minor
  - *Date of Completion*: December 2017
  - *Research Advisor*: Prof. Ryan Chornock

- **University of Toledo** – Toledo, OH USA
  - *Degree*: PhD Astrophysics
  - *Date of Completion*: July 2025
  - *Thesis Advisor*: Prof. Eli Visbal
  - *Thesis Title:* Self-consistent Theoretical Predictions of the Large-Scale Distribution of the First Stars and Galaxies

- **Stellenbosch University** – Stellenbosch, WC South Africa
  - *Position*: Post-doctoral researcher
  - *Date*: October 2025-October 2027
  - *Supervisor*: Prof. Yin-Zhe Ma

Research Interests

I am a theoretical astrophysicist and cosmologist studying the formation of the first stars and galaxies in the early Universe. I develop large-scale simulations of the early Universe with the purpose of predicting the global 21-cm signal and power spectrum at high cosmological redshifts for both standard and exotic cosmological models. I specialize in machine learning algorithms, Python & Cython programming, and have developed theoretical models spanning from simple analytic frameworks to large 3D semi-numeric simulations with self-consistent treatments for dark matter halo merger histories and radiative feedback with the goal of predicting observables for future astrophysical surveys, particularly the cosmological 21-cm signal.

Honors & Awards

- Hopkins Marching 110 Scholarship (2016-2017)
- Jayme Dyredek Fellowship Award (2017)

- Ed Foster Graduate Scholarship (2019-2020)
- Robert and Noreen Stollberg Award (2023-2024)

Positions

- Ohio University REU – May-August 2016 – Advisor: Prof. Carl Brune
    - Particle physics -- Rutherford Backscattering Spectrometry
- Particle Accelerator Operator, Ohio University – May 2016-December 2017 – Supervisor: Prof. Carl Brune
- Ohio University REU – May-August 2017 – Advisor: Prof. Ryan Chornock
    - Astrophysics -- Reduction of SNe imagery in IRAF
- Quality Lab Technician, Associated Spring Raymond, Maumee, Ohio – August 2018-August 2019 – Supervisor: Thomas J. Fischer
- Post-doctoral researcher – October 2025-October 2027 – Supervisor: Prof Yin-Zhe Ma
    - Astrophysics -- Cosmological 21-cm simulations

Publications

1. **Feathers, C.**, Kulkarni, M., Visbal, E., Hazlett, R. "A Global Semianalytic Model of the First Stars and Galaxies Including Dark Matter Halo Merger Histories", *ApJ*, 962, 1, (2024)
2. **Feathers, C.**, Kulkarni, M., Visbal, E. "From Dark Matter Minihalos to Large-Scale Radiative Feedback: A Self-Consistent 3D Simulation of the First Stars and Galaxies using Neural Networks", *JCAP*, 315, 043, (2025)
3. **Feathers, C.**, Visbal, E., Murray, S., Hazlett, R., Ma, Y. "Self-Consistent Simulations of the High-Redshift 21-cm Signal with Star Formation Calibrated to Hydrodynamic Simulations", MNRAS (accepted, 2026)

Conference Proceedings:

1. **Feathers, C.**, Kulkarni, M., Visbal, E. "From Dark Matter Minihalos to Large-Scale Radiative Feedback: A Self-Consistent 3D Simulation of the First Stars and Galaxies using Neural Networks", AAS Meeting #245, (2025)
2. **Feathers, C.**, Kulkarni, M., Visbal, E. "A Comparison of Modeling Approaches for the Cosmic Abundance of the First Stars and Galaxies", AAS Meeting #240, Vol. 54, No. 6, (2022)
3. Brune, C., **Feathers, C**., Kenady, B., Ingram, D., Deboer, R. "Absolute normalization of the $^{12}C(p, \gamma)^{13}N$ and $^{12}C(p, \gamma_0)^{14}N$ cross sections", APS Meeting, Division of Nuclear Physics and Physical Society of Japan (2023)

Teaching Experience

- Ohio University Undergraduate Physics Lab Teaching Assistant
  - January 2016-December 2017 – Supervisor: Dr. Stephen Goss
- University of Toledo Astronomy Lab Teaching Assistant
  - August 2019-May 2020 – Supervisor: Prof. J.D. Smith
  - August 2024-May 2025 – Supervisor: Heidi Westrick
    - Head Astronomy TA -- January-May 2025
- University of Toledo Physics II Lab Teaching Assistant
  - August 2021-May 2022, July 2024-May 2025 – Supervisors: Prof. Richard Irving and Prof. Song Cheng
  - Head Physics II Lab TA – July 2024-July 2025
  - Physics Tutor -- August 2019-May 2025

Professional Activities & Services

- University of Toledo Graduate Faculty Meeting Representative – August 2019-August 2024 – Supervisor: Prof. Nik Podraza
- Member of Society of Physics students – January 2015-December 2017
- Member of Phi Sigma Pi National Honor Fraternity – January 2015-December 2017
  - Parliamentarian (May 2016-May 2017)
- Member of Office of Multicultural Student Access and Retention (OMSAR) – August 2014-December 2017
- Member of Stellenbosch University Astronomy Society -- October 2025-October 2027
- Developed Honor’s Astrophysics degree program at Stellenbosch University -- Expected to be fully established in 2028
- Contributor to 21cmFAST since August 2026